

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Differential Equation

Q1

Let $y = f(x)$ is a solution of differential equation $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ and $f(0) = 0$ then $f(1)$ is equal to :

यदि $y = f(x)$ अवकल समीकरण $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ का हल है तथा $f(0) = 0$ तब $f(1)$ का मान है—

- (1) $\ln 2$ (2) $2 + \ln 2$ (3) $1 + \ln 2$ (4) $3 + \ln 2$

7th Jan Morning

Sol

(3)

$$e^y = t$$

$$e^y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - t = e^x$$

$$IF = e^{\int -1 \cdot dx} = e^{-x}$$

$$t(e^{-x}) = \int e^x \cdot e^{-x} dx$$

$$e^{y-x} = x + c$$

Put $x = 0, y = 0$ रखने पर then $c = 1$

$$e^{y-x} = x + 1$$

$$y = x + \ln(x + 1)$$

at $x = 1$ रखने पर, $y = 1 + \ln(2)$

02

Let $y(x)$ is solution of differential equation $(y^2 - x) \frac{dy}{dx} = 1$ and $y(0) = 1$, then find the value of x where curve cuts the x -axis

माना $y(x)$, अवकल समीकरण $(y^2 - x) \frac{dy}{dx} = 1$ का हल है तथा $y(0) = 1$, तब x का मान होगा, जहाँ वक्र x अक्ष को

प्रतिच्छेद करता है—

- (1) $2 - e$ (2) $2 + e$ (3) 2 (4) e

7th Jan Evening

Sol

(1)

$$\text{Sol. } \frac{dx}{dy} + x = y^2$$

$$\text{I.F.} = e^{\int 1 dy} = e^y$$

$$x \cdot e^y = \int y^2 \cdot e^y \cdot dy$$

$$= y^2 \cdot e^y - \int 2y \cdot e^y \cdot dy$$

$$\Rightarrow y^2 e^y - 2(y \cdot e^y - e^y) + c$$

$$x \cdot e^y = y^2 e^y - 2y e^y + 2e^y + C$$

$$x = y^2 - 2y + 2 + c \cdot e^{-y}$$

$$x = 0, \quad y = 1$$

$$0 = 1 - 2 + 2 + \frac{c}{e}$$

$$c = -e$$

$$y = 0, \quad x = 0 - 0 + 2 + (-e)(e^{-0})$$

$$x = 2 - e$$

03

If $y(x)$ is a solution of differential equation $\sqrt{1-x^2} \frac{dy}{dx} + \sqrt{1-y^2} = 0$, such that $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then

यदि $y(x)$, अवकल समीकरण $\sqrt{1-x^2} \frac{dy}{dx} + \sqrt{1-y^2} = 0$ का हल है तथा $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, तब

$$(1) \ y\left(\frac{1}{\sqrt{2}}\right) = -\frac{1}{\sqrt{2}} \quad (2) \ y\left(\frac{1}{\sqrt{2}}\right) = \frac{\sqrt{3}}{2} \quad (3) \ y\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} \quad (4) \ y\left(\frac{1}{2}\right) = \frac{1}{2}$$

8th Jan Morning

Sol

(3)

$$\frac{dy}{\sqrt{1-y^2}} + \frac{dx}{\sqrt{1-x^2}} = 0 \Rightarrow \sin^{-1}y + \sin^{-1}x = c$$

$$\text{At } x = \frac{1}{2}, y = \frac{\sqrt{3}}{2} \Rightarrow c = \frac{\pi}{2} \Rightarrow \sin^{-1}y = \cos^{-1}x$$

$$\text{Hence अतः } y\left(\frac{1}{\sqrt{2}}\right) = \sin\left(\cos^{-1}\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$$

04

Differential equation of $x^2 = 4b(y + b)$, where b is a parameter, is

$x^2 = 4b(y + b)$, जहाँ b एक प्राचल है, का अवकल समीकरण होगा :

$$(1) \quad x \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} + x^2$$

$$(2) \quad x \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} + x$$

$$(3) \quad x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx} + x^2$$

$$(4) \quad x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx} + 2x^2$$

8th Jan Evening

Sol

(2)

$$2x = 4by' \quad \Rightarrow \quad b = \frac{x}{2y'}$$

So. differential equation is $x^2 = \frac{2x}{y'} \cdot y + \left(\frac{x}{y'} \right)^2$

अतः अवकल समीकरण $x^2 = \frac{2x}{y'} \cdot y + \left(\frac{x}{y'} \right)^2 \Rightarrow x \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} + x$

05

$$(1+x) \frac{dy}{dx} = ((1+x)^2 + (y-3)), \text{ If } y(2) = 0 \text{ then } y(3) = ?$$

$$(1+x) \frac{dy}{dx} = ((1+x)^2 + (y-3)), \text{ यदि } y(2) = 0 \text{ तो } y(3) = ?$$

9th Jan Morning

Sol

3

$$\frac{dy}{dx} = (1+x) + \frac{(y-3)}{(1+x)}$$

$$\frac{dy}{dx} - \frac{1}{(1+x)} y = (1+x) - \frac{3}{(1+x)}$$

06

If यदि $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$, $y(1) = 1$ and तथा $y(x) = e$ then तब $x = ?$

- (1) $\frac{\sqrt{3}}{2}e$ (2) $\sqrt{3}e$ (3) $\sqrt{2}e$ (4) $\frac{e}{\sqrt{2}}$

9th Jan Evening

Sol

(2)

Put $y = vx$ रखने पर

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{vx^2}{x^2 + v^2x^2}$$

$$\Rightarrow \frac{1+v^2}{v^3} dv = -\frac{1}{x} dx$$

$$\Rightarrow \int \left(\frac{1}{v^3} + \frac{1}{v} \right) dv = \int -\frac{1}{x} dx$$

$$\Rightarrow -\frac{1}{2} \frac{1}{v^2} + \ln v = -\ln x + c$$

$$\Rightarrow -\frac{x^2}{2y^2} = -\ln y + c$$

When जब $x = 1$, $y = 1$ then तब

$$-\frac{1}{2} = c$$

$$\Rightarrow x^2 = y^2(1 + 2\ln y)$$

$$\Rightarrow x^2 = e^2(3)$$

$$\Rightarrow x = \pm \sqrt{3}e$$

So इसलिये $x = \sqrt{3}e$

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